

allowed to depend on p parameters, $\theta_1, \dots, \theta_p$. Fortunately, for successful application of maximum likelihood methods, we do not need to delve much into the computational issues or the large-sample statistical theory. Wooldridge (2002, Chapter 13) covers the theory of maximum likelihood estimation.

Least Squares

A third kind of estimator, and one that plays a major role throughout the text, is called a **least squares estimator**. We have already seen an example of least squares: the sample mean, $\bar{Y}$, is a least squares estimator of the population mean, μ . We already know $\bar{Y}$ is a method of moments estimator. What makes it a least squares estimator? It can be shown that the value of m that makes the sum of squared deviations

$$\sum_{i=1}^n (Y_i - m)^2$$

as small as possible is $m = \bar{Y}$. Showing this is not difficult, but we omit the algebra.

For some important distributions, including the normal and the Bernoulli, the sample average $\bar{Y}$ is also the maximum likelihood estimator of the population mean μ . Thus, the principles of least squares, method of moments, and maximum likelihood often result in the *same* estimator. In other cases, the estimators are similar but not identical.

C.5 Interval Estimation and Confidence Intervals

The Nature of Interval Estimation

A point estimate obtained from a particular sample does not, by itself, provide enough information for testing economic theories or for informing policy discussions. A point estimate may be the researcher's best guess at the population value, but, by its nature, it provides no information about how close the estimate is "likely" to be to the population parameter. As an example, suppose a researcher reports, on the basis of a random sample of workers, that job training grants increase hourly wage by 6.4%. How are we to know whether or not this is close to the effect in the population of workers who could have been trained? Because we do not know the population value, we cannot know how close an estimate is for a particular sample. However, we can make statements involving probabilities, and this is where interval estimation comes in.

We already know one way of assessing the uncertainty in an estimator: find its sampling standard deviation. Reporting the standard deviation of the estimator, along with the point estimate, provides some information on the accuracy of our estimate. However, even if the problem of the standard deviation's dependence on unknown population parameters is ignored, reporting the standard deviation along with the point estimate makes no direct statement about where the population value is likely to lie in relation to the estimate. This limitation is overcome by constructing a **confidence interval**.

We illustrate the concept of a confidence interval with an example. Suppose the population has a $\text{Normal}(\mu, 1)$ distribution and let $\{Y_1, \dots, Y_n\}$ be a random sample from this population. (We assume that the variance of the population is known and equal to unity for the sake of illustration; we then show what to do in the more realistic case that the variance

is unknown.) The sample average, $\bar{Y}$, has a normal distribution with mean μ and variance $1/n$: $\bar{Y} \sim \text{Normal}(\mu, 1/n)$. From this, we can standardize $\bar{Y}$, and, because the standardized version of $\bar{Y}$ has a standard normal distribution, we have

$$P\left(-1.96 < \frac{\bar{Y} - \mu}{1/\sqrt{n}} < 1.96\right) = .95.$$

The event in parentheses is identical to the event $\bar{Y} - 1.96/\sqrt{n} < \mu < \bar{Y} + 1.96/\sqrt{n}$, so

$$P(\bar{Y} - 1.96/\sqrt{n} < \mu < \bar{Y} + 1.96/\sqrt{n}) = .95.$$

C.17

Equation (C.17) is interesting because it tells us that the probability that the random interval $[\bar{Y} - 1.96/\sqrt{n}, \bar{Y} + 1.96/\sqrt{n}]$ contains the population mean μ is .95, or 95%. This information allows us to construct an *interval estimate* of μ , which is obtained by plugging in the sample outcome of the average, $\bar{y}$. Thus,

$$[\bar{y} - 1.96/\sqrt{n}, \bar{y} + 1.96/\sqrt{n}]$$

C.18

is an example of an interval estimate of μ . It is also called a 95% confidence interval. A shorthand notation for this interval is $\bar{y} \pm 1.96/\sqrt{n}$.

The confidence interval in equation (C.18) is easy to compute, once the sample data $\{y_1, y_2, \dots, y_n\}$ are observed; $\bar{y}$ is the only factor that depends on the data. For example, suppose that $n = 16$ and the average of the 16 data points is 7.3. Then, the 95% confidence interval for μ is $7.3 \pm 1.96/\sqrt{16} = 7.3 \pm .49$, which we can write in interval form as $[6.81, 7.79]$. By construction, $\bar{y} = 7.3$ is in the center of this interval.

Unlike its computation, the meaning of a confidence interval is more difficult to understand. When we say that equation (C.18) is a 95% confidence interval for μ , we mean that the *random* interval

$$[\bar{Y} - 1.96/\sqrt{n}, \bar{Y} + 1.96/\sqrt{n}]$$

C.19

contains μ with probability .95. In other words, *before* the random sample is drawn, there is a 95% chance that (C.19) contains μ . Equation (C.19) is an example of an **interval estimator**. It is a random interval, since the endpoints change with different samples.

A confidence interval is often interpreted as follows: “The probability that μ is in the interval (C.18) is .95.” This is incorrect. Once the sample has been observed and $\bar{y}$ has been computed, the limits of the confidence interval are simply numbers (6.81 and 7.79 in the example just given). The population parameter, μ , though unknown, is also just some number. Therefore, μ either is or is not in the interval (C.18) (and we will never know with certainty which is the case). Probability plays no role once the confidence interval is computed for the particular data at hand. The probabilistic interpretation comes from the fact that for 95% of all random samples, the constructed confidence interval will contain μ .

To emphasize the meaning of a confidence interval, Table C.2 contains calculations for 20 random samples (or replications) from the $\text{Normal}(2, 1)$ distribution with sample size $n = 10$. For each of the 20 samples, $\bar{y}$ is obtained, and (C.18) is computed as $\bar{y} \pm 1.96/\sqrt{10} = \bar{y} \pm .62$ (each rounded to two decimals). As you can see, the interval changes with each random sample. Nineteen of the 20 intervals contain the population value of μ . Only for replication number 19 is μ not in the confidence interval. In other words, 95% of the samples result in a confidence interval that contains μ . This did not have to be the case with only 20 replications, but it worked out that way for this particular simulation.

TABLE C.2**Simulated Confidence Intervals from a Normal($\mu, 1$) Distribution with $\mu = 2$**

Replication	$\bar{y}$	95% Interval	Contains μ ?
1	1.98	(1.36,2.60)	Yes
2	1.43	(0.81,2.05)	Yes
3	1.65	(1.03,2.27)	Yes
4	1.88	(1.26,2.50)	Yes
5	2.34	(1.72,2.96)	Yes
6	2.58	(1.96,3.20)	Yes
7	1.58	(.96,2.20)	Yes
8	2.23	(1.61,2.85)	Yes
9	1.96	(1.34,2.58)	Yes
10	2.11	(1.49,2.73)	Yes
11	2.15	(1.53,2.77)	Yes
12	1.93	(1.31,2.55)	Yes
13	2.02	(1.40,2.64)	Yes
14	2.10	(1.48,2.72)	Yes
15	2.18	(1.56,2.80)	Yes
16	2.10	(1.48,2.72)	Yes
17	1.94	(1.32,2.56)	Yes
18	2.21	(1.59,2.83)	Yes
19	1.16	(.54,1.78)	No
20	1.75	(1.13,2.37)	Yes

Confidence Intervals for the Mean from a Normally Distributed Population

The confidence interval derived in equation (C.18) helps illustrate how to construct and interpret confidence intervals. In practice, equation (C.18) is not very useful for the mean of a normal population because it assumes that the variance is known to be unity. It is easy

to extend (C.18) to the case where the standard deviation σ is known to be any value: the 95% confidence interval is

$$[\bar{y} - 1.96\sigma/\sqrt{n}, \bar{y} + 1.96\sigma/\sqrt{n}]. \quad \text{C.20}$$

Therefore, provided σ is known, a confidence interval for μ is readily constructed. To allow for unknown σ , we must use an estimate. Let

$$s = \left(\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 \right)^{1/2} \quad \text{C.21}$$

denote the sample standard deviation. Then, we obtain a confidence interval that depends entirely on the observed data by replacing σ in equation (C.20) with its estimate, s . Unfortunately, this does not preserve the 95% level of confidence because s depends on the particular sample. In other words, the random interval $[\bar{Y} \pm 1.96(S/\sqrt{n})]$ no longer contains μ with probability .95 because the constant σ has been replaced with the random variable S .

How should we proceed? Rather than using the standard normal distribution, we must rely on the t distribution. The t distribution arises from the fact that

$$\frac{\bar{Y} - \mu}{S/\sqrt{n}} \sim t_{n-1}, \quad \text{C.22}$$

where $\bar{Y}$ is the sample average and S is the sample standard deviation of the random sample $\{Y_1, \dots, Y_n\}$. We will not prove (C.22); a careful proof can be found in a variety of places [for example, Larsen and Marx (1986, Chapter 7)].

To construct a 95% confidence interval, let c denote the 97.5th percentile in the t_{n-1} distribution. In other words, c is the value such that 95% of the area in the t_{n-1} is between $-c$ and c : $P(-c < t_{n-1} < c) = .95$. (The value of c depends on the degrees of freedom $n - 1$, but we do not make this explicit.) The choice of c is illustrated in Figure C.4. Once c has been properly chosen, the random interval $[\bar{Y} - c \cdot S/\sqrt{n}, \bar{Y} + c \cdot S/\sqrt{n}]$ contains μ with probability .95. For a particular sample, the 95% confidence interval is calculated as

$$[\bar{y} - c \cdot s/\sqrt{n}, \bar{y} + c \cdot s/\sqrt{n}]. \quad \text{C.23}$$

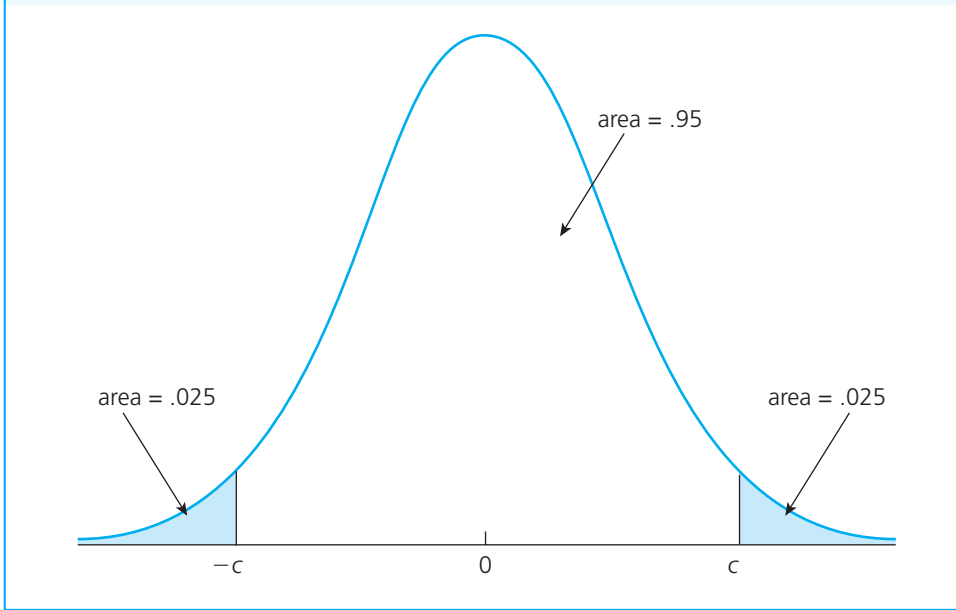
The values of c for various degrees of freedom can be obtained from Table G.2 in Appendix G. For example, if $n = 20$, so that the df is $n - 1 = 19$, then $c = 2.093$. Thus, the 95% confidence interval is $[\bar{y} \pm 2.093(s/\sqrt{20})]$, where $\bar{y}$ and s are the values obtained from the sample. Even if $s = \sigma$ (which is very unlikely), the confidence interval in (C.23) is wider than that in (C.20) because $c > 1.96$. For small degrees of freedom, (C.23) is much wider.

More generally, let c_α denote the $100(1 - \alpha)$ percentile in the t_{n-1} distribution. Then, a $100(1 - \alpha)\%$ confidence interval is obtained as

$$[\bar{y} - c_{\alpha/2} s/\sqrt{n}, \bar{y} + c_{\alpha/2} s/\sqrt{n}]. \quad \text{C.24}$$

Obtaining $c_{\alpha/2}$ requires choosing α and knowing the degrees of freedom $n - 1$; then, Table G.2 can be used. For the most part, we will concentrate on 95% confidence intervals.

There is a simple way to remember how to construct a confidence interval for the mean of a normal distribution. Recall that $\text{sd}(\bar{Y}) = \sigma/\sqrt{n}$. Thus, $s/\sqrt{n}$ is the point estimate of $\text{sd}(\bar{Y})$. The associated random variable, $S/\sqrt{n}$, is sometimes called the **standard error**

FIGURE C.4**The 97.5th percentile, c , in a t distribution.**

of $\bar{Y}$. Because what shows up in formulas is the point estimate $s/\sqrt{n}$, we define the standard error of $\bar{y}$ as $se(\bar{y}) = s/\sqrt{n}$. Then, (C.24) can be written in shorthand as

$$[\bar{y} \pm c_{\alpha/2} \cdot se(\bar{y})].$$

C.25

This equation shows why the notion of the standard error of an estimate plays an important role in econometrics.

Example C.2

[Effect of Job Training Grants on Worker Productivity]

Holzer, Block, Cheatham, and Knott (1993) studied the effects of job training grants on worker productivity by collecting information on “scrap rates” for a sample of Michigan manufacturing firms receiving job training grants in 1988. Table C.3 lists the scrap rates—measured as number of items per 100 produced that are not usable and therefore need to be scrapped—for 20 firms. Each of these firms received a job training grant in 1988; there were no grants awarded in 1987. We are interested in constructing a confidence interval for the change in the scrap rate from 1987 to 1988 for the population of all manufacturing firms that could have received grants.

We assume that the change in scrap rates has a normal distribution. Since $n = 20$, a 95% confidence interval for the mean change in scrap rates μ is $[\bar{y} \pm 2.093 \cdot se(\bar{y})]$, where $se(\bar{y}) = s/\sqrt{n}$. The value 2.093 is the 97.5th percentile in a t_{19} distribution. For the particular sample values, $\bar{y} = -1.15$ and $se(\bar{y}) = .54$ (each rounded to two decimals), so the 95% confidence interval is $[-2.28, -.02]$. The value zero is excluded from this interval, so we conclude that, with 95% confidence, the average change in scrap rates in the population is not zero.

TABLE C.3**Scrap Rates for 20 Michigan Manufacturing Firms**

Firm	1987	1988	Change
1	10	3	-7
2	1	1	0
3	6	5	-1
4	.45	.5	.05
5	1.25	1.54	.29
6	1.3	1.5	.2
7	1.06	.8	-.26
8	3	2	-1
9	8.18	.67	-7.51
10	1.67	1.17	-.5
11	.98	.51	-.47
12	1	.5	-.5
13	.45	.61	.16
14	5.03	6.7	1.67
15	8	4	-4
16	9	7	-2
17	18	19	1
18	.28	.2	-.08
19	7	5	-2
20	3.97	3.83	-.14
Average	4.38	3.23	-1.15

At this point, Example C.2 is mostly illustrative because it has some potentially serious flaws as an econometric analysis. Most importantly, it assumes that any systematic reduction in scrap rates is due to the job training grants. But many things can happen over the

course of the year to change worker productivity. From this analysis, we have no way of knowing whether the fall in average scrap rates is attributable to the job training grants or if, at least partly, some external force is responsible.

A Simple Rule of Thumb for a 95% Confidence Interval

The confidence interval in (C.25) can be computed for any sample size and any confidence level. As we saw in Section B.5, the t distribution approaches the standard normal distribution as the degrees of freedom gets large. In particular, for $\alpha = .05$, $c_{\alpha/2} \rightarrow 1.96$ as $n \rightarrow \infty$, although $c_{\alpha/2}$ is always greater than 1.96 for each n . A *rule of thumb* for an approximate 95% confidence interval is

$$[\bar{y} \pm 2 \cdot \text{se}(\bar{y})].$$

C.26

In other words, we obtain $\bar{y}$ and its standard error and then compute $\bar{y}$ plus and minus twice its standard error to obtain the confidence interval. This is slightly too wide for very large n , and it is too narrow for small n . As we can see from Example C.2, even for n as small as 20, (C.26) is in the ballpark for a 95% confidence interval for the mean from a normal distribution. This means we can get pretty close to a 95% confidence interval without having to refer to t tables.

Asymptotic Confidence Intervals for Nonnormal Populations

In some applications, the population is clearly nonnormal. A leading case is the Bernoulli distribution, where the random variable takes on only the values zero and one. In other cases, the nonnormal population has no standard distribution. This does not matter, provided the sample size is sufficiently large for the central limit theorem to give a good approximation for the distribution of the sample average $\bar{Y}$. For large n , an *approximate* 95% confidence interval is

$$[\bar{y} \pm 1.96 \cdot \text{se}(\bar{y})],$$

C.27

where the value 1.96 is the 97.5th percentile in the standard normal distribution. Mechanically, computing an approximate confidence interval does not differ from the normal case. A slight difference is that the number multiplying the standard error comes from the standard normal distribution, rather than the t distribution, because we are using asymptotics. Because the t distribution approaches the standard normal as the df increases, equation (C.25) is also perfectly legitimate as an approximate 95% interval; some prefer this to (C.27) because the former is exact for normal populations.

Example C.3

[Race Discrimination in Hiring]

The Urban Institute conducted a study in 1988 in Washington, D.C., to examine the extent of race discrimination in hiring. Five pairs of people interviewed for several jobs. In each pair, one person was black and the other person was white. They were given résumés indicating that they were virtually the same in terms of experience, education, and other factors that determine job qualification. The idea was to make individuals as similar as possible with the exception of race. Each person in a pair interviewed for the same job, and the researchers recorded which applicant received a job offer.

This is an example of a *matched pairs analysis*, where each trial consists of data on two people (or two firms, two-cities, and-so-on) that are thought to be similar in many respects but different in one important characteristic.

Let θ_B denote the probability that the black person is offered a job and let θ_W be the probability that the white person is offered a job. We are primarily interested in the difference, $\theta_B - \theta_W$. Let B_i denote a Bernoulli variable equal to one if the black person gets a job offer from employer i , and zero otherwise. Similarly, $W_i = 1$ if the white person gets a job offer from employer i , and zero otherwise. Pooling across the five pairs of people, there were a total of $n = 241$ trials (pairs of interviews with employers). Unbiased estimators of θ_B and θ_W are $\bar{B}$ and $\bar{W}$, the fractions of interviews for which blacks and whites were offered jobs, respectively.

To put this into the framework of computing a confidence interval for a population mean, define a new variable $Y_i = B_i - W_i$. Now, Y_i can take on three values: -1 if the black person did not get the job but the white person did, 0 if both people either did or did not get the job, and 1 if the black person got the job and the white person did not. Then, $\mu \equiv E(Y_i) = E(B_i) - E(W_i) = \theta_B - \theta_W$.

The distribution of Y_i is certainly not normal—it is discrete and takes on only three values. Nevertheless, an approximate confidence interval for $\theta_B - \theta_W$ can be obtained by using large sample methods.

Using the 241 observed data points, $\bar{b} = .224$ and $\bar{w} = .357$, so $\bar{y} = .224 - .357 = -.133$. Thus, 22.4% of black applicants were offered jobs, while 35.7% of white applicants were offered jobs. This is *prima facie* evidence of discrimination against blacks, but we can learn much more by computing a confidence interval for μ . To compute an approximate 95% confidence interval, we need the sample standard deviation. This turns out to be $s = .482$ [using equation (C.21)]. Using (C.27), we obtain a 95% CI for $\mu = \theta_B - \theta_W$ as $-.133 \pm 1.96(.482/\sqrt{241}) = -.133 \pm .031 = [-.164, -.102]$. The approximate 99% CI is $-.133 \pm 2.58(.482/\sqrt{241}) = [-.213, -.053]$. Naturally, this contains a wider range of values than the 95% CI. But even the 99% CI does not contain the value zero. Thus, we are very confident that the population difference $\theta_B - \theta_W$ is not zero.

Before we turn to hypothesis testing, it is useful to review the various population and sample quantities that measure the spreads in the population distributions and the sampling distributions of the estimators. These quantities appear often in statistical analysis, and extensions of them are important for the regression analysis in the main text. The quantity σ is the (unknown) population standard deviation; it is a measure of the spread in the distribution of Y . When we divide σ by $\sqrt{n}$, we obtain the **sampling standard deviation** of $\bar{Y}$ (the sample average). While σ is a fixed feature of the population, $\text{sd}(\bar{Y}) = \sigma/\sqrt{n}$ shrinks to zero as $n \rightarrow \infty$: our estimator of μ gets more and more precise as the sample size grows.

The estimate of σ for a particular sample, s , is called the sample standard deviation because it is obtained from the sample. (We also call the underlying random variable, S , which changes across different samples, the sample standard deviation.) Like $\bar{y}$ as an estimate of μ , s is our “best guess” at σ given the sample at hand. The quantity $s/\sqrt{n}$ is what we call the standard error of $\bar{y}$, and it is our best estimate of $\sigma/\sqrt{n}$. Confidence intervals for the population parameter μ depend directly on $\text{se}(\bar{y}) = s/\sqrt{n}$. Because this standard error shrinks to zero as the sample size grows, a larger sample size generally means a smaller confidence interval. Thus, we see clearly that one benefit of more data is that they result in narrower confidence intervals. The notion of the standard error of an estimate, which in the vast majority of cases shrinks to zero at the rate $1/\sqrt{n}$, plays a fundamental role in hypothesis testing (as we will see in the next section) and for confidence intervals and testing in the context of multiple regression (as discussed in Chapter 4).